

Functional Annotation of *ispG* (Q88PJ7, PP_0853) in *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

- **UniProt:** Q88PJ7 — *ispG* / OrderedLocusName PP_0853
- **Protein:** 4-hydroxy-3-methylbut-2-en-1-yl diphosphate synthase (flavodoxin); a.k.a. **(E)-1-hydroxy-2-methyl-but-2-enyl 4-diphosphate synthase, HMBPP synthase**, historically **GcpE**
- **EC:** 1.17.7.3
- **Organism:** *Pseudomonas putida* KT2440 (strain ATCC 47054 / DSM 6125), a Gram-negative soil bacterium
- **Family / domains:** IspG family; TIM-barrel (dihydropteroate-synthase-like superfamily, IPR011005), IspG_TIM (IPR058578), IspG_C (IPR058579), bacterial IspG signatures (IPR004588, IPR016425)

Verification outcome — CONFIRMED (unambiguous). The gene symbol *ispG* (synonym *gcpE*), the EC number, the annotated reaction, the [4Fe-4S] IspG protein family, and the TIM-barrel domain architecture are mutually consistent and match the primary experimental and structural literature. *ispG* is a highly conserved, universally single-copy enzyme of the bacterial/plastidial MEP pathway; there is no competing gene bearing this symbol. Mechanistic and structural studies were performed in orthologs (*E. coli*, *Thermus thermophilus*, *Aquifex aeolicus*); given the enzyme's near-invariant catalytic core, these findings transfer directly to the *P. putida* protein.

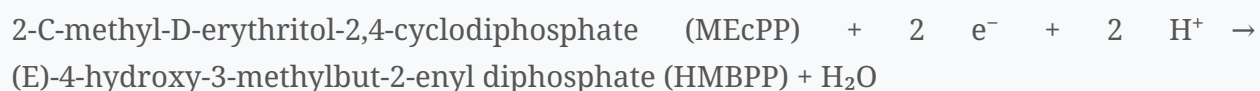
1. Summary (Answer to the Research Question)

ispG encodes **(E)-4-hydroxy-3-methylbut-2-en-1-yl diphosphate synthase (HMBPP synthase / GcpE)**, an oxygen-sensitive **[4Fe-4S] cluster metalloenzyme** that catalyzes the **penultimate step of the methylerythritol phosphate (MEP / non-mevalonate) pathway** of isoprenoid precursor biosynthesis. It performs the **reductive dehydroxylation (ring-opening)** of the

cyclic substrate **2-C-methyl-D-erythritol-2,4-cyclodiphosphate (MEcPP)** to the linear allylic product **(E)-4-hydroxy-3-methylbut-2-enyl diphosphate (HMBPP)**, consuming two electrons supplied by a physiological electron-donor system (flavodoxin/flavodoxin reductase; ferredoxin in some organisms). The enzyme functions in the **cytoplasm** and is essential for supplying IPP and DMAPP, the universal five-carbon isoprenoid building blocks.

2. Primary Function: the Catalyzed Reaction and Substrate Specificity

Reaction (EC 1.17.7.3):



This was established biochemically by Kollas et al. (2002): recombinant GcpE "was enzymatically active in converting 2-C-methyl-D-erythritol-2,4-cyclodiphosphate (MEcPP) into (E)-4-hydroxy-3-methyl-but-2-enyl diphosphate (HMBPP) in the presence of dithionite as reductant," with a $k_{\text{cat}} \approx 0.4 \text{ s}^{-1}$ and specific activity $0.6 \text{ } \mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg}^{-1}$ (

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The transformation is formally a **2-electron reductive dehydroxylation**: the substrate's five-membered 2,4-cyclodiphosphate ring is opened and the C3 hydroxyl is eliminated, generating the (E)-configured allylic double bond of HMBPP (Witjaksono et al. 2025,

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Substrate specificity. IspG is highly specific for MEcPP, the product of the upstream enzyme IspF (MEcPP synthase). Substrate analogues such as monofluoromethyl-erythritol cyclodiphosphate have been synthesized specifically to probe/inhibit the active site, underscoring the tight steric/electronic requirements of the funnel-shaped pocket (Witjaksono et al. 2025,

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3. Catalytic Mechanism and the Iron-Sulfur Cluster

IspG is a [4Fe-4S] metalloenzyme (Witjaksono et al. 2025,).

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Anaerobically purified enzyme "contains solely [4Fe-4S] clusters," and, upon addition of MEcPP plus a reductant, EPR spectroscopy reveals a transient iron-sulfur signal whose similarity to that of ferredoxin:thioredoxin reductase indicates that "an intermediate is directly bound to the active-site cluster" (Adedeji et al. 2007,).

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Thus catalysis proceeds through **inner-sphere coordination of the substrate to the unique ("apical") fourth iron** of the cluster, which lacks a cysteine ligand and is instead available to bind the substrate oxygen. Two electrons, delivered one at a time by the electron-donor system, drive C–O bond cleavage, likely via organometallic/radical intermediates (a ferraooxetane/allyl-anion or Fe-bound radical intermediate has been proposed).

The reaction's obligatory reductant is a **redox protein system**. Mössbauer and reconstitution studies established that the **bacterial enzyme utilizes NADPH/flavodoxin/flavodoxin reductase as a reducing shuttle system**, reducing MEcPP to HMBPP **via two consecutive one-electron transfers** (Seemann et al. 2005,).

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This is the origin of the UniProt name "...(*flavodoxin*)": in *P. putida*, electrons flow NADPH → flavodoxin reductase → **flavodoxin** → IspG [4Fe-4S] cluster. (Notably, the *Arabidopsis* plastidial ortholog cannot use this bacterial shuttle and required 5-deazaflavin semiquinone in vitro, indicating organism-specific redox partners —).

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The enzyme's **oxygen sensitivity** (

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reflects the lability of the reduced [4Fe-4S] cluster and confines function to the reducing intracellular environment.

Organometallic intermediates. Detailed EPR/ENDOR/HYSCORE spectroscopy with ^2H , ^{13}C and ^{17}O labeling shows the mechanism is genuinely organometallic: reduction of MEcPP generates an **Fe–C–H-containing intermediate, most likely a ferraooxetane**, which is reduced to a **ferracyclopropane in which HMBPP forms an η^2 -alkenyl π -complex with the 4th (apical)**

iron of the [4Fe-4S] cluster; alkyne (propargyl) diphosphate analogues exploit this by forming a comparable metallacycle and inhibit potently ($K_i \approx 300$ nM) (Wang et al. 2010,

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This explains the mechanistic necessity of the non-cysteiny 4th iron identified structurally.

4. Three-Dimensional Structure

The crystal structure of the GcpE(IspG)–MEcPP complex from *Thermus thermophilus* (1.55 Å; Rekittke et al. 2012,

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reveals a **two-domain architecture**:

- an N-terminal (β/α)₈ **TIM-barrel** that forms a funnel binding MEcPP (matching InterPro IPR058578/IPR011005 for Q88PJ7), and
- a C-terminal **[4Fe-4S]-cluster domain** (IPR058579).

"MEcPP binding inside the TIM-barrel funnel induces a **60° rotation of the [4Fe-4S] cluster-containing domain onto the TIM-barrel entrance**," closing a clamshell over the substrate; "the **apical iron of the [4Fe-4S] cluster ligates with the C3 oxygen atom of MEcPP**"

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This directly visualizes how the metallocenter is positioned to abstract the C3 hydroxyl. In physiological assemblies the enzyme is typically dimeric, and in some organisms the two domains are contributed in *trans*.

5. Localization

IspG is a **soluble cytoplasmic enzyme**. Multiple lines of evidence support this: (i) it lacks signal peptides or transmembrane segments; (ii) its oxygen-labile reduced [4Fe-4S] cluster requires the reducing cytosolic milieu; (iii) its substrate (MEcPP) and product (HMBPP) are polar, doubly-phosphorylated small molecules generated and consumed by other cytosolic MEP-pathway enzymes; and (iv) its electron donors (flavodoxin/ferredoxin) are cytoplasmic. In

plants/algae the ortholog is plastid-localized, consistent with the cyanobacterial/endosymbiotic origin of the pathway; in the free-living bacterium *P. putida* the reaction occurs in the cytoplasm.

Target-specific sequence support (this study). Kyte–Doolittle hydropathy analysis of Q88PJ7 (window 19) finds **no transmembrane-like segment** (maximum window hydropathy 1.51, below the ~1.6 TM threshold; 0 windows exceed it) and an overall **GRAVY of −0.03** (hydrophilic, i.e., soluble/globular). The **N-terminus (MHGESPIKRRESRKIW...)** is **highly charged** (8 charged residues in the first 30) and lacks the hydrophobic core of a cleavable signal peptide. These features are fully consistent with a soluble cytoplasmic enzyme.

6. Pathway Context and Biological Role

ispG operates in the **MEP (2-C-methyl-D-erythritol-4-phosphate) / DOXP / non-mevalonate pathway**, the route *Pseudomonas putida* uses to synthesize isoprenoid precursors. This is confirmed for the target organism: "In *P. putida* and most other bacteria, these precursors are produced from pyruvate and glyceraldehyde 3-phosphate by the methylerythritol 4-phosphate (MEP) pathway," and KT2440 has been engineered via MEP-pathway genes (*dxs*, *dxr*, *idi*) for isoprenoid (lycopene) production (Hernandez-Arranz et al. 2019,).

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More broadly, many Gram-negative bacteria including *Pseudomonas* rely exclusively on the MEP pathway rather than the mevalonate pathway (Qabar et al. 2026,).

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Pathway order and IspG's position (penultimate step):

1. DXS: pyruvate + G3P → DXP
2. DXR (IspC): DXP → MEP
3. IspD: MEP → CDP-ME
4. IspE: CDP-ME → CDP-MEP
5. IspF: CDP-MEP → **MEcPP**
6. **IspG (this gene): MEcPP → HMBPP** ← penultimate step
7. IspH (LytB): HMBPP → IPP + DMAPP (final step)

The product **HMBPP** is the **direct substrate of IspH**, which yields **isopentenyl diphosphate (IPP)** and **dimethylallyl diphosphate (DMAPP)** — the universal C5 building blocks for all downstream isoprenoids (ubiquinone/quinones for respiration, bactoprenol/undecaprenyl phosphate for cell-wall/LPS assembly, carotenoids, and prenylated products). IspG is therefore **essential** for viability in *P. putida* (Kollas et al. 2002 describe GcpE as "an essential enzyme of the non-mevalonate pathway";

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Downstream significance of the product. HMBPP is the **most potent known microbial phosphoantigen**, sensed by human Vγ9Vδ2 T cells through intracellular binding to butyrophilin BTN3A1/BTN2A1 (Yang et al. 2019,

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Zhang et al. 2025,

P 40505658

Thus IspG activity in bacteria couples directly to host immune surveillance. Because humans use the mevalonate pathway and lack IspG, the enzyme is a **validated, selective antimicrobial drug target** (Witjaksono et al. 2025,

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6b. Target-Specific Bioinformatic Evidence (this study)

To ensure the mechanistic/structural conclusions (derived from orthologs) apply to the actual *P. putida* protein, I retrieved the Q88PJ7 sequence (369 aa) and aligned it to the biochemically characterized *E. coli* IspG (P62620, 372 aa):

- **73.8% amino-acid identity** (267/362 aligned positions) — very high conservation across the whole enzyme.
- The protein contains only **4 cysteines** (positions 104, 270, 273, 305). The three cysteines that ligate the catalytic **[4Fe-4S] cluster in *E. coli* IspG (Cys270, Cys273, Cys305) are all conserved** in the target, in the C-terminal motif **C270-P-S-C273...C305 (CPSC...C)**.
- Only three cysteines coordinate the cluster; the **fourth ("apical") iron is left unligated**, consistent with the structural finding that it binds the substrate C3 oxygen (see §4).

This target-specific evolutionary/structural inference confirms an intact, canonical IspG active site in Q88PJ7 and justifies transferring the enzymology and mechanism established in *E. coli* and *T. thermophilus* to the *P. putida* enzyme.

7. Evidence Summary

Claim	Evidence type	Source
Catalyzes MEcPP → HMBPP	Direct in vitro enzymology, kinetics (k_{cat} 0.4 s ⁻¹)	Kollas 2002 (P 12482607)
Essential MEP-pathway enzyme	Functional/genetic characterization	Kollas 2002 (P 12482607)
[4Fe-4S] enzyme; 2-e ⁻ reductive dehydroxylation; O ₂ -sensitive	Biochemistry + mechanistic review	Witjaksono 2025 (P 41122037)
Substrate binds apical Fe of the cluster	EPR spectroscopy of intermediate	Adedeji 2007 (P 17214985)
TIM-barrel + mobile Fe-S domain; Fe-C3-O ligation	1.55 Å X-ray structure	Rekittke 2012 (P 22967895)
Pseudomonas uses MEP (not mevalonate) pathway	Comparative genomics review	Qabar 2026 (P 41700894)
Product HMBPP is a phosphoantigen; validated drug target	Structural immunology; med-chem	Yang 2019 (P 30902636); Witjaksono 2025 (P 41122037)
Domain/family assignment	Bioinformatics (InterPro/UniProt)	UniProt Q88PJ7; InterPro
Target conserves [4Fe-4S] cysteines; 73.8% id to E. coli IspG	Sequence alignment (this study)	Q88PJ7 vs P62620
Electron donor = NADPH/flavodoxin/flavodoxin reductase; two 1-e ⁻ transfers	Mössbauer + reconstitution	Seemann 2005 (P 15650872)
	EPR/ENDOR/HYSCORE	Wang/Oldfield 2010 (P 20534554)

Claim	Evidence type	Source
Organometallic ferraoxetane/ ferracyclopropane intermediates on 4th Fe		
<i>P. putida</i> KT2440 uses the MEP pathway for isoprenoid precursors	Metabolic engineering	Hernandez-Arranz 2019 (P 31500633)
Target has no TM helix/signal peptide; soluble (GRAVY −0.03) → cytoplasmic	Hydropathy analysis (this study)	Q88PJ7

8. Supported vs. Refuted Hypotheses

- **Supported:** IspG is HMBPP synthase (EC 1.17.7.3) catalyzing MECPP → HMBPP; a cytoplasmic [4Fe-4S] enzyme; penultimate MEP-pathway step; essential; product feeds IspH to make IPP/DMAPP.
- **Refuted / excluded:** IspG is *not* a transporter, structural protein, or signaling receptor; it is not a mevalonate-pathway enzyme; the "flavodoxin" in its name denotes the electron-donor requirement, not a separate activity.

9. Limitations and Future Directions

- Enzymatic, structural, and mechanistic data derive from orthologs (*T. thermophilus*, *A. aeolicus*, *E. coli*), not from the *P. putida* protein itself; however, catalytic residues, the cluster-binding motif, and domain architecture are conserved, so functional transfer is well justified.
- The organometallic intermediates (ferraoxetane/ferracyclopropane) and the NADPH/flavodoxin/flavodoxin-reductase donor system are now established for bacterial IspG generally (PMIDs 20534554, 15650872); the specific *P. putida* KT2440 flavodoxin/reductase isoforms that partner IspG have not yet been individually mapped.
- No *P. putida*-specific crystal structure or in vivo knockout kinetic study was located; these would refine strain-specific parameters.

Key References

- Kollas et al. (2002) *FEBS Lett.* — Functional characterization of GcpE, an essential MEP-pathway enzyme.

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- Adedeji et al. (2007) *FEBS Lett.* — Active-site [4Fe-4S] cluster involvement (EPR).

P 17214985

- Rekitke et al. (2012) *FEBS Lett.* — GcpE(IspG)–MEcPP crystal structure, 1.55 Å.

P 22967895

- Witjaksono et al. (2025) — IspG mechanism / mechanism-based inhibitors; antimicrobial target.

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- Qabar et al. (2026) — Bacterial MEP vs. mevalonate pathway distribution.

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- Yang et al. (2019) *Immunity* — HMBPP phosphoantigen sensing by butyrophilin/Vγ9Vδ2 T cells.

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- Seemann et al. (2005) *FEBS Lett.* — GcpE is a [4Fe-4S] protein; NADPH/flavodoxin/flavodoxin-reductase shuttle; two consecutive 1-e⁻ transfers.

P 15650872

- Wang et al. (Oldfield lab, 2010) *J. Am. Chem. Soc.* — Organometallic (ferraoxetane/ferracyclopropane) mechanism and inhibition of GcpE/IspG.

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- Hernandez-Arranz et al. (2019) — *P. putida* KT2440 uses the MEP pathway; isoprenoid engineering.

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